



OBJECT ORIENTED SOFTWARE ENGINEERING

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Extreme Programming (XP) approach

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Other Agile Frameworks include:

1. **Extreme Programming**
2. **Kanban**
3. **DevOps**

- **XP** is the **most prominent** Agile Software development method.
- **Prescribes** a set of **daily stakeholder practices**.
- “**Extreme**” levels of practicing leads to more **responsive software**.
- Changes are more **realistic, natural, inescapable**.

XP Values

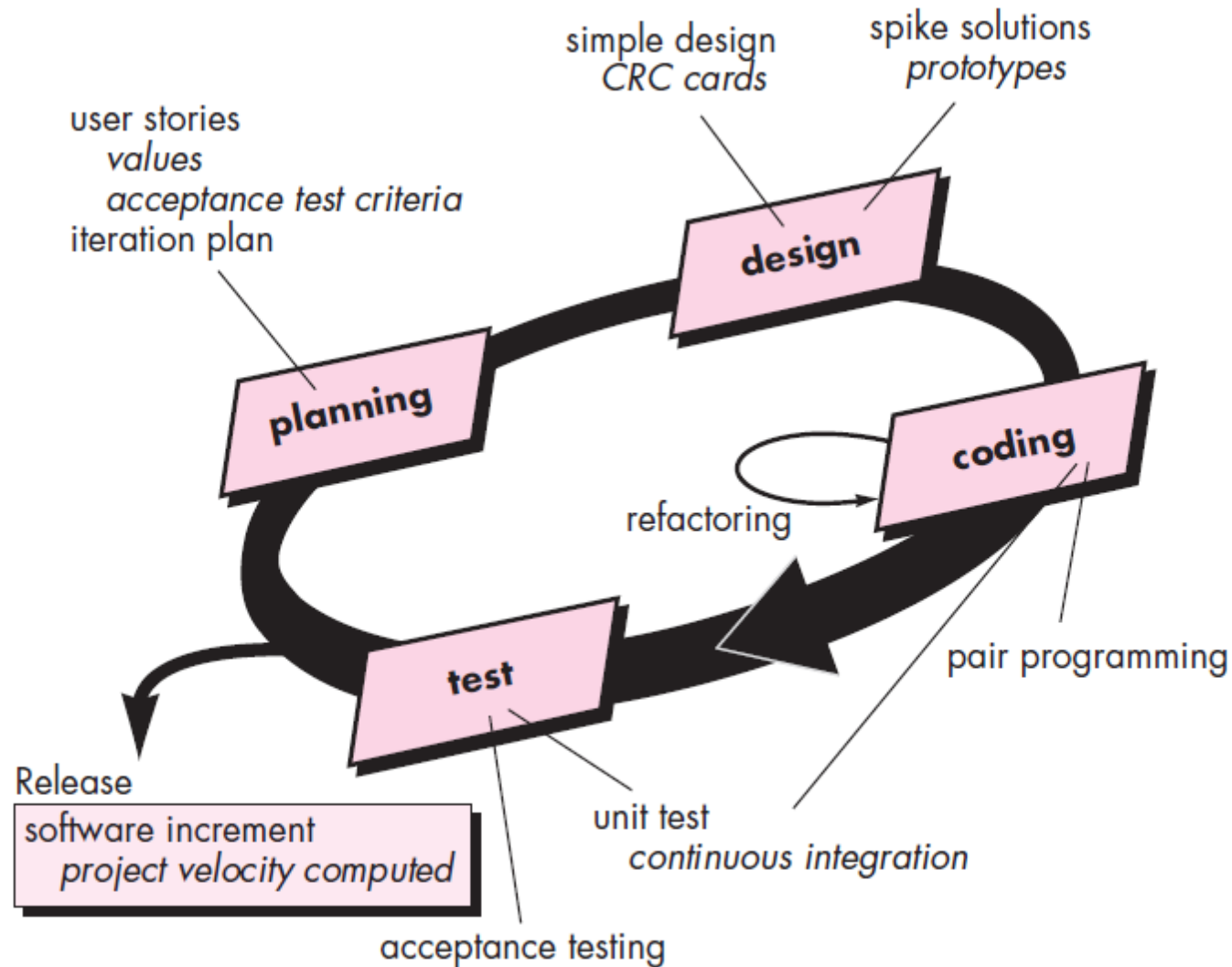
- A set of **five values** that establish a foundation for all work performed as part of XP—
 1. **Communication,**
 2. **Simplicity,**
 3. **Feedback,**
 4. **Courage, and**
 5. **Respect**
- Each of these values is used as **a driver** for specific XP activities, actions, and tasks.

XP Process

- The most widely used agile process, originally proposed by **Kent Beck**.
- Extreme Programming uses an **object-oriented approach**.
- Encompasses a **set of rules and practices** that occur within the context of **four** framework activities: **planning, design, coding, and testing**.

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1. Extreme Programming (XP) - XP Process



1. Extreme Programming (XP) – XP Process

1. XP Planning:

- The planning activity *begins with **listening*** , Listening leads to the creation of a set of “**stories**” *that describe **required output, features, and functionality*** for software to be built.
 - Begins with the creation of “**user stories**”.
 - Agile team assesses each story and assigns a **cost**.
 - Stories are grouped for a **deliverable increment**.
 - A **commitment** is made on delivery date.
 - After the first increment “**project velocity**” is used to help define subsequent delivery dates for other increments.

2. XP Design:

- Encourage the use of **CRC cards** as an **effective mechanism** for thinking about the software in an **object-oriented context**.
- CRC (class-responsibility-collaborator) cards **identify and organize the object-oriented classes** that are relevant to the current software increment.

2. XP Design...

Class-Responsibility Collaborator(CRC).

Class: <i>Librarian</i>	
Responsibilities	Collaborators
<i>check in book</i>	<i>Book</i>
<i>check out book</i>	<i>Book, Borrower</i>
<i>search for book</i>	<i>Book</i>
<i>knows all books</i>	
<i>search for borrower</i>	<i>Borrower</i>
<i>knows all borrowers</i>	

Class CardReader	
Responsibilities	Collaborators
Tell ATM when card is inserted	ATM
Read information from card	Card
Eject card	
Retain card	

1. Extreme Programming (XP) – XP Process...

2. XP Design...

- For difficult design problems, XP suggests the creation of “**spike solutions**” — a design prototype.
- A spike solution is a very simple program to explore potential solution.
- **Difficult design** should be modeled using **prototype**.

1. Extreme Programming (XP)

3. XP Coding:

- Recommends the **construction of a unit test** for a story *before coding commences*. This helps developer to better focus on what must be implemented to pass the test.
- Encourages “**pair programming**”. This provides a mechanism for real time problem solving, real time quality assurance as it is created.
- As pair programmers complete their work, the code they develop is integrated with the work of other.
- All **unit tests** are executed daily

1. Extreme Programming (XP)

3. XP Coding...

- XP encourages “**refactoring**”—an iterative refinement of the internal program design.
 - Refactoring is a process of changing a software system in such a way that **it does not alter the external behavior** of the code, **yet improves the internal structure.**

4. XP Testing:

- All **unit tests** are **executed daily**.
- “**Acceptance tests**” are defined by the customer and executed to assess customer visible functionality.
 - XP *acceptance tests*, also called **customer tests**, are specified by the customer and focus on **overall system features and functionality** that are visible and reviewable by the customer.



THANK YOU

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